

Technical Document #800: Retrieval Augmented Generation - Comprehensive Overview

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.
- Hybrid search combines keyword-based and semantic search.